

Design and Implementation of an IoT-based Smart Home with the Ability to Communicate with the Smart Grid

Amin Hajjafari

Department of Electrical Engineering
Shahid Beheshti University
Tehran, Iran
aminhajjafari30@gmail.com

Reza Badiei

Department of Electrical Engineering
Shahid Beheshti University
Tehran, Iran
kiabad78@gmail.com

Hadi Asharioun

Faculty of Electrical Engineering
Department
Shahid Beheshti University
Tehran, Iran
asharioun@sbu.ac.ir

Ramak Nasiri

Department of Computer Engineering
Alzahra University
Tehran, Iran
ramaknasiri@gmail.com

Abstract—Nowadays, although smartization has been widely accepted, the communication between a smart home and the smart grid has not been given enough attention. This means monitoring and checking the smart grid orders in the peak power consumption. Thus, we have designed and implemented a smart home based on the internet of things, which is capable to be in touch with the smart grid for observation and ordering. In this project, we have presented an architecture based on IoT in which all the sections are under observation and control. In addition, electricity consumption is being measured all the time and the information is collected from the smart grid through the smart power meter. This project has been installed in the IoT laboratory at Shahid Beheshti University and the result was acceptable.

Keywords— *smart grid, IoT, smart home, device, gateway, server*

I. INTRODUCTION

A smart home is a modern house or building where all its electrical components and appliances are controlled by a single centrally integrated system. All smart home components are connected to each other through this central system, and they work well together. In a smart home, individuals can even control all the electrical components and appliances of the house wherever they are.

This project was the construction and complete installation of a smart house on the Internet of Things platform.

Electricity as the primary source of power for gadgets, computers, machinery, and other electronic goods is used

at the home-to-industrial level. Most of the world's Electric Grids operate on fossil fuels which are getting thinner and expected to run out soon, the best solution that is possible to do is Smart Grid. The Smart Grid realized extensive use of information sensing, transmission and processing. Today, IoT technology become an essential role in construction of power grid. This paper reviews the Integration of IoT in Smart Grids. IoT-integrated Smart Grid systems are already deployed, but the full capabilities of instant knowledge and sustainable large-scale data processing have not been exploited optimally. Two of the architectures of IoT integrated Smart Grids is four-layered architecture and Web Enabled Smart Grid. To process and manage the Big Data in Smart Grid, Supervisory Control and Data Acquisition (SCADA) system can be used and furthermore enhanced with cloud computing. The smart grid is the future of electric grid that solves the problems of information flows, rising energy consumption, energy waste, reliability and security in the conventional power grid.[1]

Nowadays, energy crisis is one of the most important issues faced by most countries. Given the accommodation of a large population, high-rise buildings have a significant role in creating or resolving this crisis. A recent solution with regard to the optimization and reduction of energy consumption is using smart systems in buildings. In fact, with the help of modern knowledge, smart buildings consume energy in the right place and time. By transforming a simple building into a dynamic one, not only will it be able to adapt to changing environmental conditions, it will also consider the living habits of dwellers and comfort standards in order to provide maximum satisfaction. Moreover, the money spent on making smart

appliances will be fully compensated after a short while, saving the overall costs and energy.[2]

The smart grid gives us an output that is related to our power consumption. As a result, by analyzing this output, we can understand the consumption load and control it if needed.

In houses, smartization is manifested in two ways: automatic home management and remote control, which are very useful in reducing and controlling energy and power consumption. The most appropriate platform for implementing smart ambience in homes to create a suitable network in the scientific and commercial field is the use of the Internet of Things which has been discussed in this project.

Bringing intelligence into our houses gives us more abilities about controlling and observing the condition of the house. Accordingly, one important impact is that a smart house is more likely to be able to reduce the costs and expenses anyway, in addition, it will be useful to save more energy with due attention to the problems of producing energy in today's world.

From another aspect, having intelligent houses can give us more abilities with wider capabilities to use. This has brought us more information about different parts of our houses and various actions available. This can be used by the user of the house in different ways and the more important part is the automatic orders which can control the house sans the need for an observer.

Before the advent of IoT, communication was only human-to-human and human-to-things; Nowadays, IoT has made communication between things possible. The Internet of Things provides communication between devices through the network. Our network usually consists of small networks in which devices communicate with each other and then these small networks are connected to a larger one.

The Internet of Things (IoT) is an emerging paradigm that enables communication between electronic devices and sensors through the internet to facilitate our lives. IoT uses smart devices and the internet to provide innovative solutions to various challenges and issues related to various business, governmental and public/private industries across the world [3].

Smart home technology, also often referred to as home automation or domotics (from the Latin "Domus" meaning home), provides homeowners security, comfort, convenience and energy efficiency by allowing them to control smart devices, often by their smartphone or other networked device [4].

The main purpose of this project is to implement a smart home and all its expected features on the platform of IoT. Making suitable devices for smart home on the platform of the Internet of Things, such as registration system devices with presence and absence capability to open the door using fingerprint, password, tag, and face

detection. Making other devices such as electrical switches, temperature sensor, etc. Moreover, creating the necessary conditions and parameters for the implementation of more capabilities in the future, such as advanced artificial intelligence and data analysis which are the main components of this project.

II. ARCHITECTURE

A. Main structure

As understood, the main system is divided into three parts. At first, the devices are at the front of the system, they have two main duties which are sensing and actuating. They sense the information and then actuate following the order they receive with due attention to the sensed data. second, there is a Gateway which is the most key part of an IoT platform. It has many duties but the most important one is to be a communication bridge between the first part and the third part. Furthermore, Gateway helps the devices to communicate with each other, so it works as the center of the network and connects all the parts to each other. The third is the server working as the brain of our system. The server's responsibility is to collect devices' data from the gateway and then use them for operations such as storing and processing, then it sends back orders to the gateway to control devices. These orders could be from the intelligent system decider or the user's decisions through the dashboard platform.

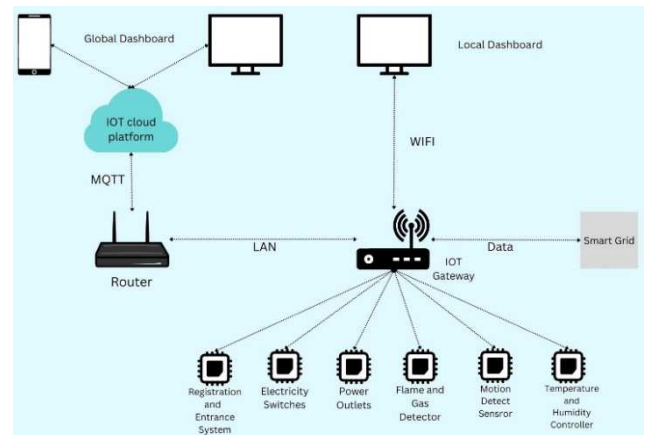


Figure 1. The diagram illustrates the main architecture of the IoT smart home system.

B. Devices

Devices are the most variable part of the smart house. The basic abilities of the system are being built here, because the more different sensors or actuators we have, the more capability we can benefit in our smart house. In making devices three pillar parts have got to be worked on well, the microprocessor, sensors and actuators.

Esp8266 has been chosen as the microprocessor of the devices, this is because of its physical abilities, power and low price. This microprocessor has been programmed using the "Arduino IDE" compiler. the next important part

is the sensors and actuators connected to the processor, sensors detect information from the external environment and give them to the processor, on the other hand, actuators receive data about the detail of what they must do from the microprocessor. This data transfer happens through the GPIO pins with different data transfer protocols like Serial and SPI. With different sensors and actuators, we can make a variety of devices or even combine them on a single device. In this project, many objects became smart in the house, some of them have been mentioned below.

Power Outlets: All the power outlets are equipped with a device that checks the power usage of them and informs the gateway.

Electricity Switches: Switches could be controlled by the gateway, additionally they could be controlled manually using a touch sensor installed on their frame. The flow is being controlled using a relay which makes switches and

Registration and Entrance System: As it was said, this system contains 5 different ways for registration, face detection, fingerprint, tag, password and Bluetooth. When someone is registered, then the device sends a message to the gateway to inform the person and asks the entrance system, which is a switch installed on the door to open the door.

Temperature and Humidity Controller: This device comprises of a temperature and humidity sensor (DHT11) and a seven-segment. Therefore, the information is sent to the gateway every single minute and it is available on the seven-segment for the user in the house.

Motion Detect Sensor: This device is responsible to notify the gateway whether the room is empty or not, this has been made possible using a PIR sensor (SR602) placed on the frame of the device. Whenever the device finds out the room has been clear for a couple of minutes, it sends the related message to the gateway. In addition, immediately after the room gets filled, the device informs the gateway again about the condition. This device could also be used as a burglar alarm.

Flame and Gas Detector: A gas (MQ135) and a flame sensor have been placed on this device and whenever they detect fire or smoke, the device will keep notifying the gateway till the flame, or the smoke is gone.

Two sensors are common in all devices. To know the energy consumption in each device, they are equipped with a current (ACS712) and a voltage (ZMPT101B) sensor; therefore, the instantaneous current and voltage in every single device are being measured. With this information, it is possible to find out the energy being used in all parts of the house at any time. The information about the voltage and current of each device should be sent to the server to calculate the energy being used in any part of the house and make decisions about them.

The power supply used in the devices is an AC-to-DC converting city electricity to a 5V DC voltage supply, a 5V-to-3V regulator also has been prepared for modules working with 3V voltage of VCC. All the stuff of a device has been initiated on a PCB sheet designed by Altium Designer application to have better quality and smaller size of the device at the same time.

C. Gateway

Depending on the duties of the gateway, different processors and microprocessors could be used. In this project, we decided to choose a powerful processor to be able to give more tasks to it, so the decision was Raspberry Pi 3b board. This board contains Wi-Fi, Bluetooth, an Ethernet module, and a powerful quad-core 64-bit CPU with 1GB RAM. The Gateway's operating system is decided to be Raspbian to run the Raspberry Pi. Furthermore, the application used for organizing the gateway to do all the tasks in internal and external networks is Node-Red, which helps to make easier connections likewise making programming orders.

Gateway's first responsibility is to communicate with devices in the internal network and with the server through the external network referred to [6][7]. Another responsibility of the gateway is to connect to the wireless cameras in the house and send the video signal to the server.

Our gateway is connected to the Internet through its Ethernet module using a LAN connector, additionally, it uses wi-fi to give local access to the users for observing the house condition and giving orders via a web dashboard.

D. Communication with the smart grid

An important goal of this smart house is to communicate with the smart grid (in case the power meter). In addition to the flow connections that the smart grid makes with the consumers, it contains a data connection that consists of information about the energy consumption of the house. This information is the exact energy usage of the house in short periods of time. This data comes to the smart power meter of the house and gateway collects it from the smart power meter. Then gateway sends the collected information to the server for process and actions.

E. Server

The server is responsible to do most of the smart house's process. There are two main objects defined for the server, the first is to collect the information coming from the gateway and do some operations like saving information and data analysis, second is to develop a web page for manual control and orders. Then whether the order is from the intelligent system or the user, it sends the order to the gateway for actuation.

The data coming to the server from the gateway could be different from each other and the server must analyze them and make decisions. We have two types of

information, first is from devices that arrive occasionally about the events happening in the house, and second is the data and orders about the smart grid and energy consumption of devices. The server contains an Artificial Intelligence system that helps it to order. This system has two algorithms for intentions, first one is about energy consumption, it checks the information about devices' energy consumption and smart grid, then according to them if electricity usage is more than the demand it sends orders to the gateway to decrease the usage as much as we need. Then there are some rules for making an automatic controlling system which makes operations instead of the user of the house, for example, it turns off the lights in the morning, whenever the room is empty it lowers the energy consumption and brings back everything to its initial state when someone enters as well.

In a smart home equipped with Artificial Intelligence, energy consumption and carbon emissions are minimized through the automatic control of smart thermostats, smart outlets and smart lighting systems. The accuracy of a suitable quantized LSTM-based (long short-term memory- a variant of RNN) deep-learning model is higher than 90%. The delay for user authentication is very small. Besides, RNNs are persuasive lightweight choices for convolutional neural networks (CNNs) for many sensor devices [8]. Hence, this algorithm has been used in the AI part of this project.

To store the data MySQL was the database management system on the server referred to [9]. Then some python programs based on Artificial Intelligence use the received information to make automatic decisions. besides, this information is available on the web page for user observation.

To create the web page, we need two frameworks for the frontend and backend processes referred to [10][11]. React JS has been chosen as the frontend framework. The quality of this framework in creating dynamic pages makes this framework a proper choice for this project. The backend framework is decided to be Express JS as it has an appropriate non-blocking IO system because it works based on Node JS as the programming language. Another benefit of these two frameworks is that they both work with Java Script language which makes them easier to work with at the same time. Using this system, users can have access to the information about the house in real-time, moreover, the users will be able to order using their computer or smartphone.

III. COMMUNICATION AND PROTOCOLS

The internal network is based on a RF wireless transceiver (RFM69). All the devices and the gateway use this specific transceiver to communicate with each other through our protocol which will be explained in the following. There are other protocols for IoT systems that are ready to use but we preferred to design our protocol to make it more compatible with the needs of a smart home,

also we will be able to make updates and changes the way it will be needed in the future. Encrypting the messages is one important thing that has got to be worked on in the future.

The external network works with the internet. The gateway connects to the internet through the router in the house and makes communication with the server using MQTT protocols through a broker [12] installed on the server.

As it was said, the gateway is the center of the communications whether for the internal or external network and it will be used as the communication bridge between devices and the smart grid on one side and the server on the other side. All the messages from the server to devices or vice versa should contain eight parts to be valid. First, we need to have a pass to avoid being interrupted by other desultory messages which are not about our system, so when we receive a message, this part could work like a signature of our system in the message to be recognized. After that, the next part has to determine which house are we talking about, when the number of smart houses number increase in the future, this segment will be a very crucial one in our network. Third, the type of the device should be characterized, so it will be understandable if the message is about one of the switches or power outlets, etc. After understanding the type of device, the next section is to define which specific device is the message about. As we found out the exact device, the element after that shows the side of the message whether the destination is the server or one of the devices. Because the messages should be counted and we need to avoid repetitive messages, the sixth part is responsible to count the messages. The seventh section must determine the type of the message, in other words, the subject of the content should be selected here, for example, if it is an order for actuation or a request for a status announcement. The last section is the message content; the main substance will be placed after all addresses mentioned before.

Table I. The message protocol's arrangement for the system's inner communication

1	2	3	4	5	6	7	8
Pass	House Number	Device Type	Device Number	Side	Message Number	Message Type	Content

Every single device checks the first six fragments to use the information if it was for itself, but if the message did not match its demands, then a mesh protocol system will be activated and repeats the message and helps it to get to its destination.

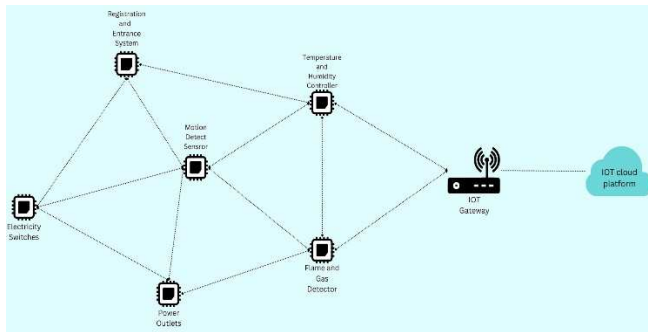


Figure 2. The mesh protocol diagram which shows the connection between gateway and devices.

IV. RESULT

Eventually, all the mentioned devices in the architecture section have been installed and tested in the IoT lab in the Electrical Engineering faculty at Shahid Beheshti University in Tehran and they are utilized several times a day by people who use that room.



Figure 2. Some of the devices installed and being used as the result of the project.

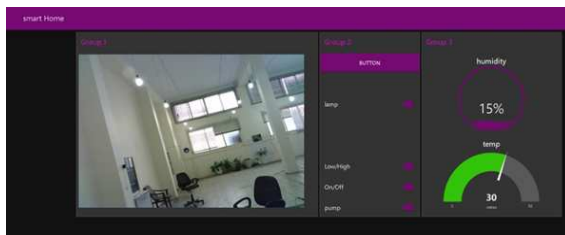


Figure 3. The designed web dashboard of the smart home, used for control and monitoring.

V. CONCLUSION

In this paper, the design of a smart house has been explained which is helpful to make our houses work properly and easier to control. The condition in the house should be sensed by the sensing devices placed in the house and they must send the information to the gateway. Thus, the gateway collects the information and sends them again to the server. Then the server has two responsibilities, first, it analyzes the received data and makes decisions, also it has to create a user interface for manual orders from any distance through the internet. Eventually, the same steps should be taken back; therefore, the actuator devices

receive orders and make actions to control the house conditions. This gives us better performance, wider abilities, and most importantly the capability to reduce energy usage daily.

REFERENCES

- [1]A,Pramudhita.,R,Asmara.,I,Siradjuddin.,E,Rohadi.(2018).,"Internatio
nal Conference on Applied science and technology(iCasT)",p2.
- [2]M,Danesh.,J Abdolhosseini.(2021)."International Journal of Civil
Engineering'.p1.
- [3]S,Kumar., P,Tiwari., M,Zimblar.(2019)." Internet of Things is a
revolutionary approach for future technology enhancement: a
review", Journal of Big Data.p1.
- [4]V,Kumar., R,Kumar Chawda." International Journal of Engineering
Applied Sciences and Technology".(2020).p530.
- [5]V,Kumar., R,Kumar Chawda." International Journal of Engineering
Applied Sciences and Technology".(2020).p2.
- [6]C. I. Ramos Incháustegui, F. Rodríguez and S. Gutiérrez,
"Development and Testing of Gateway LoRa for Cloudino IoT Open
Source Platform," 2020 IEEE ANDESCON, 2020, pp. 1-5.
- [7]B. Oniga, A. Munteanu and V. Dadarlat, "Open-source multi-protocol
gateway for Internet of Things," 2018 17th RoEduNet Conference:
Networking in Education and Research (RoEduNet), 2018, pp. 1-6.
- [8]S,Sepasgozar,R,Karimi,L,Farahzadi,F,Moezzi.(2020)."ResearchGate".
p23.
- [9]P. T. Anh Mai, J. K. Nurminen and M. Di Francesco, "Cloud Databases
for Internet-of-Things Data," 2014 IEEE International Conference on
Internet of Things (iThings), and IEEE Green Computing and
Communications (GreenCom) and IEEE Cyber, Physical and Social
Computing (CPSCoM), 2014, pp. 117-124.
- [10]K,Arto,T,Ahola,T,Vatienan(2016)."From the front end of projects to
the back end of operations: Managing projects for value creation
throughout the system lifecycle",International Journal of Project
Management p258-270.
- [11]Ismet Isnaini, & Muhammad Naufal Shidqi (2020). Preliminary
design of website application for environmental and radiation
monitoring using react. In Chevy CahyanaGede Bayu Suparta, Gede
Bayu Suparta, & Abdul Mun'im (Eds.). Proceedings of the National
Seminar on Innovation and Utilization of Nuclear Technology 2020,
(p. 569). Indonesia: National Nuclear Energy Agency.
- [12]K. Grgić, I. Špeh and I. Hedi, "A web-based IoT solution for
monitoring data using MQTT protocol," 2016 International
Conference on Smart Systems and Technologies (SST), 2016, pp.
249-2